

# Patient Informatics



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# Outline



- Lecture learning outcomes
- Introduction
- The Internet for Patient Medical Education
- Reasons for the increased use of the Internet in the healthcare arena
- Patient Web Portals
- Personal health records
- Patient-physician communication via secure e-mail and e-visits

# Lecture Learning Outcomes

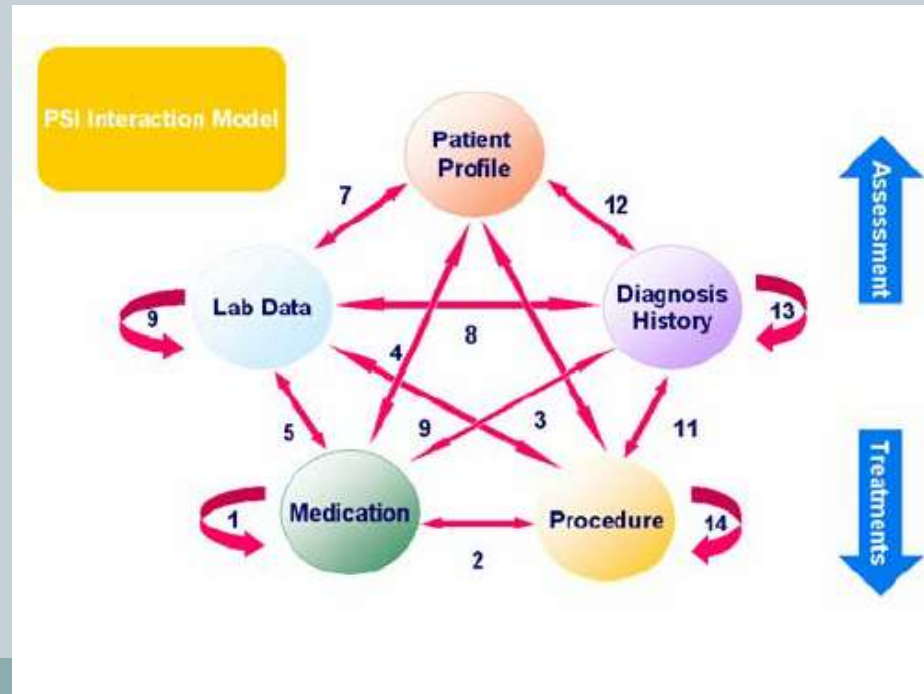


- Outline the features of electronic patient information sources and the various ways of communication between patients and healthcare provider
- Explain the role of patient informatics in a healthcare system

# Introduction



- Patient Informatics (also known as consumer informatics) is a new aspect of Medical Informatics that largely reflects the empowered healthcare consumer.



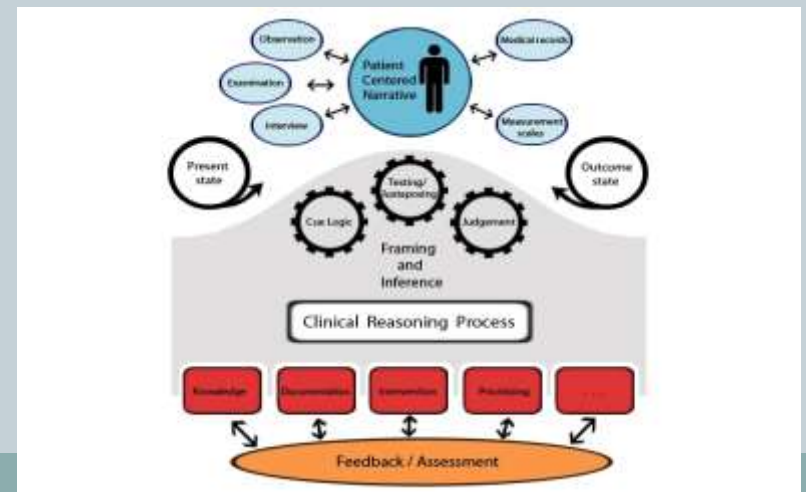
# Topics in Patient Informatics



- 1. The Internet for patient medical education
- 2. Web portals for patient access to their healthcare system
- 3. Personal health records
- 4. Patient-physician communication via secure e-mail and e-visits

# I. The Internet for Patient Medical Education

- Adults have broadband access at home or at work.
- The individuals most likely to seek health related information online are women, age younger than 65, college graduates, those with more online experience and those with broadband access.



|                         |  |
|-------------------------|--|
| Topics Searched         |  |
| Specific Disease        |  |
| Medical Treatment       |  |
| Diet and Nutrition      |  |
| Exercise                |  |
| Medication Issues       |  |
| Alternative Medications |  |
| Insurance Companies     |  |
| Depression              |  |
| Doctor or Hospital      |  |

# Reasons for the Increased Use of the Internet in the Healthcare Arena



- Healthcare is becoming more patient-centered in general.
- Quality patient education web sites abound
- Most non-medical businesses are offering an online method to promote services, pay bills, etc
- Patients are becoming impatient about finding the right answers, the best physicians, the best hospitals and the best medical care at the lowest cost
- Patients expect their physicians to be tech savvy and would consider switching if they are not.
- Healthcare organizations are using information technology as a marketing tool to attract patients
- Searching the Internet is now very fast and easy

# Examples of Online Patient Education Web Sites

- WebMD.
- Revolution Health.
- MedlinePlus.
- Healthfinder.
- FamilyDoctor.



# Medline Plus

- 700+ health topics
- Drug information
- Health encyclopedia
- 165 tutorials
- Videos of surgical procedures
- Topics available in 40 languages



## Health Topics

Find information on health, wellness, disorders and conditions



## Drugs & Supplements

Learn about prescription drugs, over-the-counter medicines, herbs, and supplements



## Videos & Tools

Discover tutorials, health and surgery videos, games, and quizzes



## Medical Encyclopedia

Articles and images for diseases, symptoms, tests, treatments

## Medical Dictionary from Merriam-Webster

Look up in Dictionary

GO



## Senior Fitness

Learn the best ways to get active and stay active on our [Exercise for Seniors](#) page.

1 2 3 4 II

## Today's Health News

[Dietary Mineral Could Be One Key to Blood Pressure Control](#)

[Do More to Fight HIV in Africa: Study](#)

[More Older Americans Cared for At Home](#)

[More health news](#)

## Clinical Trials

Search [ClinicalTrials.gov](#) for drug and treatment studies.

## Stay Connected

Sign up for MedlinePlus email updates

Enter email address

GO

## NIH MedlinePlus Magazine

Read the [latest issue](#)

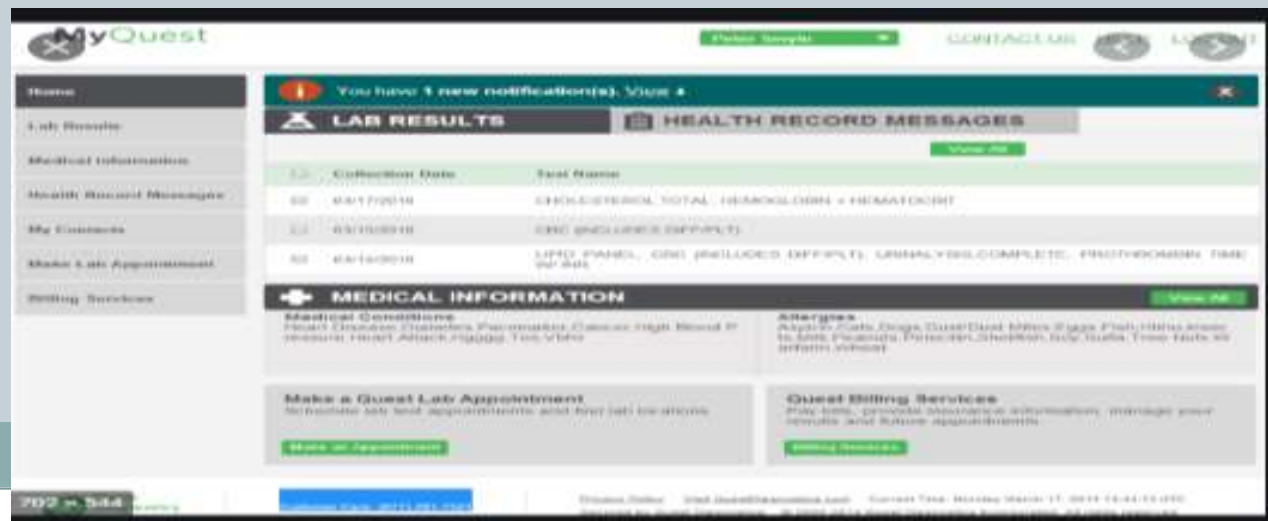


- Health check tools: quizzes, calculators and self-assessments
- Health dictionary
- Directories to locate physicians and hospitals
- Other health databases
- Health news

## II. Patient Web Portals



- Web portals are web-based programs that patients can access for health related services.
- A web portal can be a standalone program or it can be integrated with an electronic health record.
- Patient portals began as a web based entrance to a healthcare system for the purpose of learning about a hospital, healthcare system or physician's practice. Also, they were clearly a marketing ploy to attract patients who were Internet savvy.



# Patient Services (Web Portal Features and comments)

| Feature                        | Comments                                                                                            |
|--------------------------------|-----------------------------------------------------------------------------------------------------|
| Online registration            | Allows patients to complete information before a visit or hospitalization                           |
| Medication refills             | Secure message can be left for physician to refill or renew medications, instead of telephone calls |
| Laboratory results             | Patients can find results on recent tests as well as an explanation                                 |
| Electronic visits              | Portals exist that facilitate e-visits and the payment process                                      |
| Patient education              | Links to common educational sites                                                                   |
| Personal health records (PHRs) | Allows patients and their families to create and update their PHR                                   |
| Online appointments            | Allows patients to see what appointments are available and when                                     |
| Referrals                      | Patients can request referrals to specialists, e.g. OB-GYN                                          |
| Secure messaging               | More convenient than playing phone tag                                                              |
| Bill paying                    | Online payment using credit card is faster than snail mail                                          |
| Document uploading             | Several portals allow uploading of medical records to their site                                    |
| Tracking function              | Portal allows patients to upload diet, blood sugars, blood pressures, etc                           |

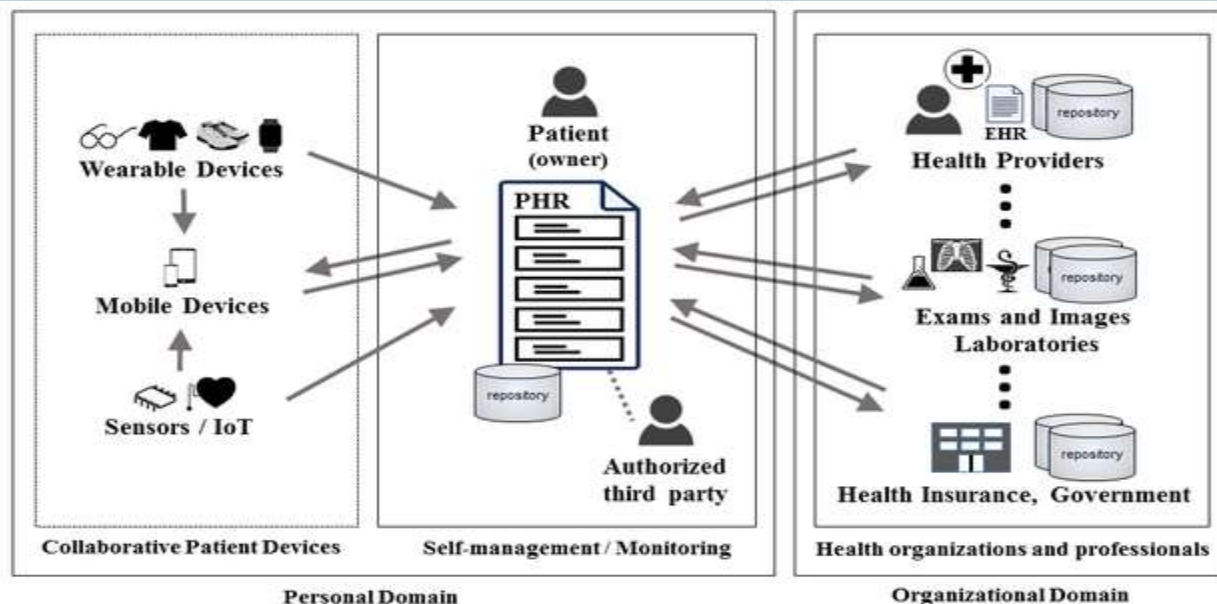
# Examples of Patient Web Portals

- MySaintAls.
- Epic MyChart
- RelayHealth.



# III. Personal Health Records (PHRs)

- “An electronic, universally available, lifelong resource of health information needed by individuals to make health decisions”.



# Ideal PHR Features



- Portable, meaning that the information will follow you even when there is a job, insurer or clinician change.
- Interoperable, meaning the PHR format can be shared among disparate partners.
- Auto populated, which would mean that clinical and test results would be inputted automatically
- Controlled by the patient
- Longitudinal record and not just a snapshot
- Private and secure
- Must fit into clinician's workflow and not be a separate process



# PHR Formats



- Tethered- connected to one platform and not interoperable
- Untethered PHRs- not connected to one platform and there have more interoperability potential

# Untethered PHRs



- Web based.
- Mobile technology- Patient information can be downloaded to: Secure digital cards and USB drives, Smart Phones, Smart Cards.
- Examples: Google Health, Microsoft HealthVault

The screenshot shows the Google Health interface for a user named Jane. At the top, there's a search bar and a 'Search the Web' button. Below the user's name, there are links for 'Options', 'Print', 'Share', and 'Private'. The user's profile information is displayed: Age: 42 years old, Sex: Female, Race / Ethnicity: White, Blood type: O+. There are 'Edit' and 'Summary' tabs. The 'Summary' tab is active, showing a 'Wellness' section with a list of health metrics and their values over time. The metrics include Blood pressure (118 / 74 mmHg), Cholesterol, total (194 mg/dL), Coffee consumption (2 cups/day), Steps taken (10014 steps), and Weight (with BMI) (144.5 lb, 22 BMI). Each metric has a small line graph showing the trend. There are also 'Problems' and 'Add' buttons at the bottom.

| Metric             | Value            | Date         |
|--------------------|------------------|--------------|
| Blood pressure     | 118 / 74 mmHg    | Aug 15, 2010 |
| Cholesterol, total | 194 mg/dL        | Jul 12, 2010 |
| Coffee consumption | 2 cups/day       | Aug 15, 2010 |
| Steps taken        | 10014 steps      | Sep 7, 2010  |
| Weight (with BMI)  | 144.5 lb, 22 BMI | Sep 12, 2010 |

The screenshot shows the Microsoft HealthVault interface for a user named Quantifier. The interface is divided into several sections: 'Conditions', 'Files', 'Health History', 'Measurements', 'Medications', and 'Personal Profile'. The 'Conditions' section lists various health conditions. The 'Files' section lists various files. The 'Health History' section lists various health events. The 'Measurements' section lists various measurements. The 'Medications' section lists various medications. The 'Personal Profile' section lists various personal information. The interface is designed to be comprehensive and easy to navigate.

## IV. Forms of Patient Physician Communication and Interaction

1. Secure patient-physician e-mail communication
2. Electronic visits
3. Telephonic visits
4. Audio-Video Televisits



# Secure patient-physician e-mail communication



- Multiple studies suggest patients want to communicate with their physicians via e-mail but that enthusiasm is not shared by most physicians





- Physicians cite the following reasons for not using e-mail: liability, poor compensation, privacy issues, staff not trained and the feeling that face-to-face visits are better.



# Benefits of e-mail communication



- The communication is asynchronous so physicians can answer at their convenience and avoid “phone tag”.
- Overhead is lower for electronic messages compared to phone messages and they are self-documenting.
- Patients tend to lose less time from work for minor issues.
- Faster than telephonic communication and resulted in high consumer satisfaction.



# Disadvantages of e-mail messaging



- Potential disadvantages of e-mail messaging might include: indigent patients less likely to use the service; inability to examine the patient; potential for communication errors; possible slow responses; security issues and the potential to be overwhelmed.



# Electronic Visits



Electronic visits (e-visits or virtual visits) are an example of telehealth or telemedicine where medical care is delivered remotely.

The web-based choices for patients might include:

- Secure messaging and templates to input a health concern and wait for a response by a clinician
- Telephonic communication (audio) to communicate with clinicians
- Audio and video to state a concern and show, example, a rash



- Secure messaging: E-visits require secure messaging and not routine e-mail. This has the advantage of much better security and privacy and the ability to have a third party involved in the billing process.





- Telephonic visits: Example: TelaDoc is a telephone based consult service. Web based application guarantees a clinician will return a phone call in 3 hours with average charge.
- The clinician will prescribe and handle refills but not prescribe narcotics or order labs.





- **Audio-Video Televisits: Example: American Well.**  
Patients can interact with clinicians using web-based videoconferencing, as well as secure chat and telephonic communication.



# References



- Chapter 6, Medical Informatics: Practical Guide for the Healthcare Professional , 3rd Edition



Thank You